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PALAEOMAGNETISM OF THE MIOCENE JANTETELCO GRANODIORITES AND TEPEXCO VOLCANIC GROUP AND INFERENCES FOR CRUSTAL BLOCK ROTATIONS IN CENTRAL MEXICO

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ABSTRACT

Urrutia-Fucugauchi, J., 1981. Palaeomagnetism of the Miocene Jantetelco Granodiorites and Tepexco Volcanic Group and inferences for crustal block rotations in central Mexico. Tectonophysics, 76: 149—168.

Palaeomagnetic directions isolated from seven sites in three bodies of the Miocene Jantetelco Granodiorites and from two sites in two lava flows of the Miocene Tepexco Volcanic Group of Central Mexico point about 50° west of the expected Miocene and Middle Tertiary geomagnetic field directions for the area. Both alternating field and thermal demagnetization experiments indicate that these remanent magnetizations are stable and free from any significant secondary overprinting. The granodiorites present both normal and reverse polarity and at one site there is evidence of the coexistence of both polarities, suggesting that the cooling of the intrusive spanned at least one polarity transition of the geomagnetic field. The lavas present reverse polarity. The angular deflection is attributed to the tectonic rotation of a crustal block about a pivot located somewhere nearby. The tectonic rotation is likely related to lateral shear acting along the trace of the Mexican Volcanic Belt. The mechanism may have involved rotations of rigid crustal fragments while the area was translated in response to lateral shear along large regional bounding faults. Alternatively, the rotations may have been associated with the formation of pull-apart basins, again in conjuction with the regional lateral shear.

INTRODUCTION

Major portions of Mexico are covered by thick sequences of igneous rocks. One of these portions, which comprises the historical and present-day magmatic activity, is the Mexican Volcanic Belt (MVB). This belt traverses central Mexico from the Pacific to the Gulf of Mexico (Fig. 1), roughly between the

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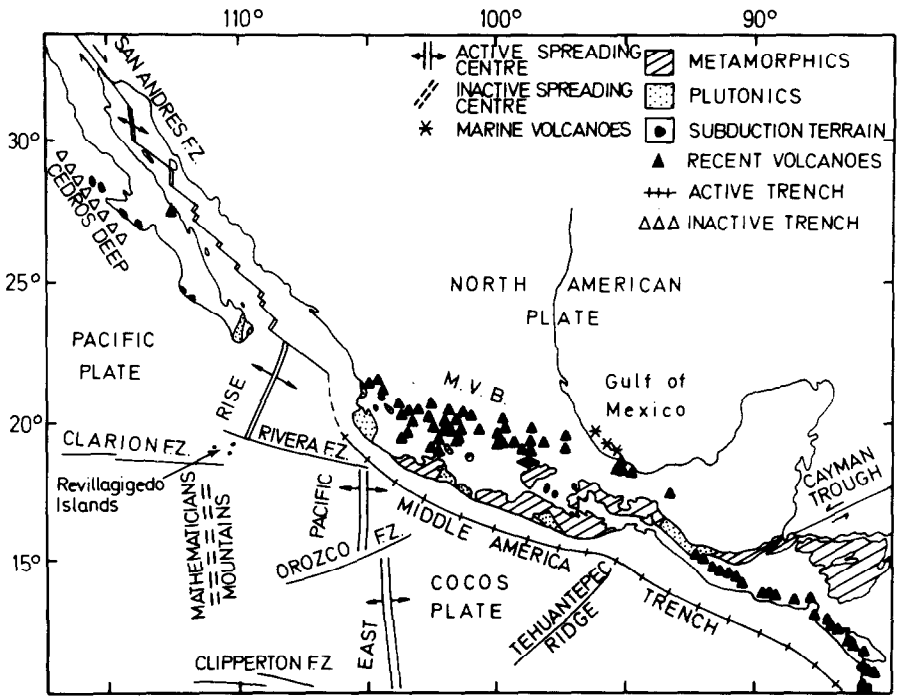


Fig. 1. Generalized schematic map of Middle America showing some of the major tectonic features. Area studied is indicated by ◆.

19°N and 21°N parallels, and includes the highest elevations of the country. Some workers have interpreted large scale tectonic movements along the trace of the belt in Late Mesozoic and Tertiary times (e.g. Gastil and Jensky, 1973; Pal and Urrutia-Fucugauchi, 1977). Earlier palaeomagnetic investigations have also indicated the possibility of relative tectonic movements along the MVB (summarized in Urrutia-Fucugauchi, 1979b), but except for some results from northeast Jalisco and the Valley of Mexico within the belt, the results come from northern Mexico and very likely all are from north to the zone where the possible movements took place. In an attempt to clarify this problem, a preliminary palaeomagnetic study of some Miocene igneous rocks from an area in central Mexico which lies at the boundary of the MVB with yet another volcanic province, the Sierra Madre del Sur, was undertaken. This work reports on the palaeomagnetic data obtained and briefly discusses the tectonic implications.

LOCATION AND GEOLOGIC SETTING

The area studied is located between about 18°36′N and 18°45′N and about 98°44′W and 98°50′W around Jonacatepec village (Fig. 2). The study

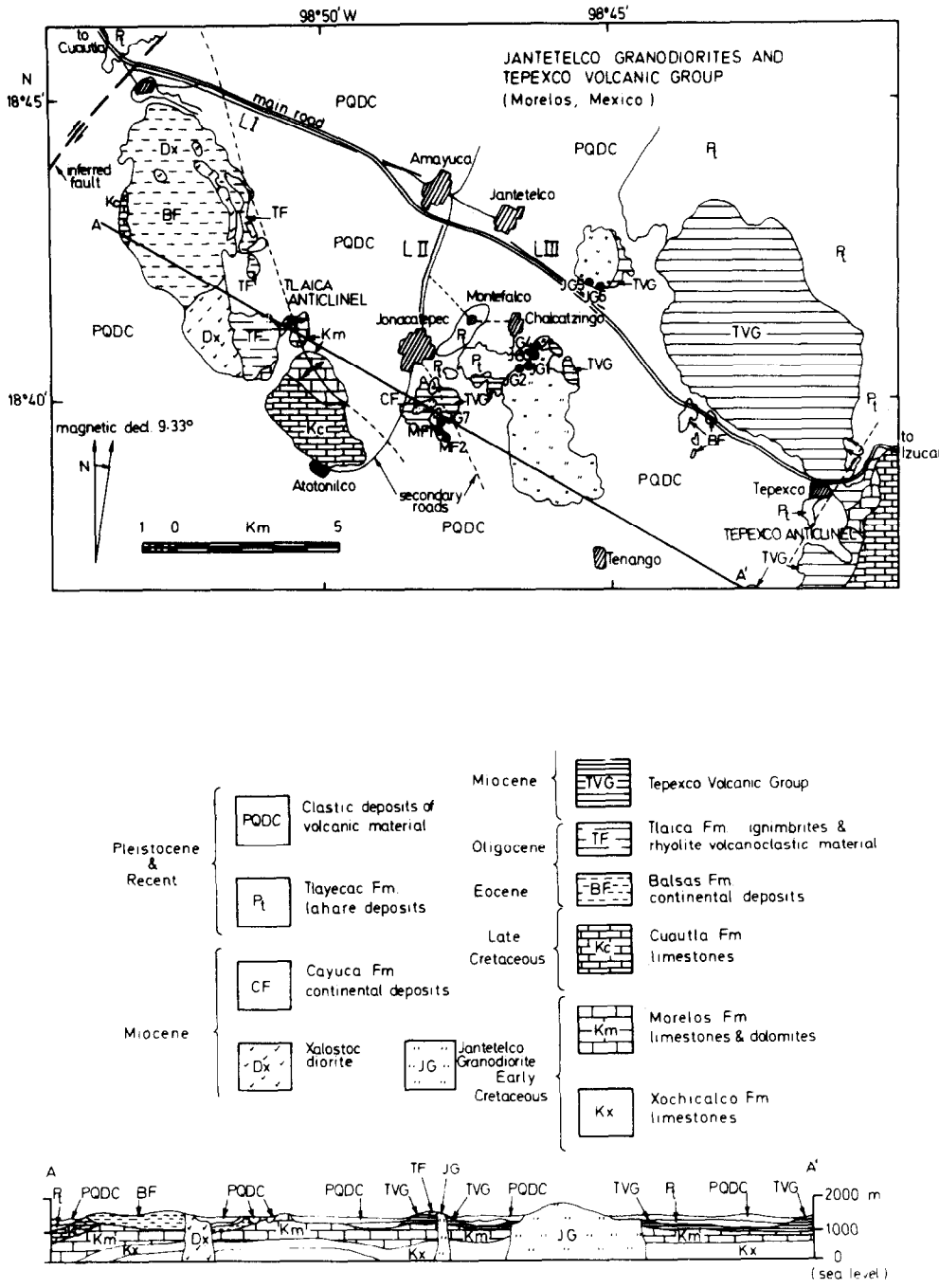


Fig. 2. Geology of the sampling area (after Fries, 1965). Sites are indicated by JG, Jantetelco Granodiorites and MF, Tepexco Volcanic Group.

was restricted to the Jantetelco Granodiorites (JG) and the Tepexco Volcanic Group (MF).

The oldest unit exposed in the area is the Morelos Formation of Albian age which is followed by two other limestone formations, the Cuautla Formation of Turonian age and the Mezcala Formation of Canocian—Campanian age. These are followed by the Late Eocene—early Late Oligocene Balsas Formation which lies in angular unconformity on the Mezcala Formation and is interpreted as a continental molasse deposit (De Cserna, 1965). This is conformably overlain by rhyolites and latites of the Tlaica Formation. To the west of this area, the igneous rocks are assigned to the Tilzapotla Formation which has been dated by the Pb-α method on zircon separates giving an age of 26 Ma (Jaffe et al., 1959). The age of the Tlaica—Tilzapotla Formation is considered Late Oligocene (Fries, 1965). This Formation is covered by rhyodacitic, dacitic and andesitic lava flows and volcanoclastic deposits of the Tepexco Volcanic Group, which are assigned to the Miocene (Fries, 1965; De Cserna, 1965). The next unit formed by lacustrine deposits, and of more limited aereal extent, is the Cayuca Formation of probable Late Miocene age. In some areas the Tepexco Group and Cayuca Formation are pierced by intrusive bodies, the Tlaica (Xalostoc) Diorites and the Jonacatepec (Chalcatzingo) Granodiorites, which are assigned to the Miocene (Fries, 1965; De Cserna, 1965). The following units are the volcanic complex of the Popocatepetl volcano and the Tlayecac Formation of Pliocene to Pleistocene age. Finally, most of the surface is covered by Recent volcanoclastic deposits and alluvium.

METHODS AND SAMPLING DETAILS

A total of 50 samples from 9 sites were collected from three bodies of the Jantetelco Granodiorites (38 samples, 7 sites) and two lavas of the Tepexco Volcanic Group (12 samples, 2 sites). The site locations are given in Fig. 2; the fourth small granodioritic body indicated in the geologic map (Fries, 1965) was not identified during the field work. A total of 96 specimens of 2.5 cm diameter and 2.2—2.3 cm height were cut from the cores. The direction and intensity of natural remanent magnetization (NRM) were measured using a Digico spinner magnetometer (Molyneux, 1971) and the axial initial susceptibilities were determined with a low-field susceptibility bridge (Collinson et al., 1963). Results from one sample (2 specimens) which was possibly misoriented in the field were rejected. Two or three pilot specimens per site with an average NRM direction and intensity for the site were selected; one was submitted to thermal demagnetization in 6—7 steps up to 600°C using a non-inductive electric furnace (Stephenson, 1967) and the others were submitted to alternating field (AF) demagnetization in 10—11 steps up to 900 Oe using an electronically controlled AF demagnetizer (De Sa and Widdowson, 1974). The optimum treatments were selected by using stability index analysis (Tarling and Symons, 1967; Symons and Stupavsky, 1974) and vec-

tor substraction analysis (Zijderveld, 1967). Except for results from one site, the results corresponded to well defined stable end points. In conjunction with these measurements, isothermal remanent magnetization (IRM) and anhysteretic remanent magnetization (ARM) acquisition experiments, AF demagnetization of IRM and ARM, and viscous effect analysis were performed on some samples to investigate the magnetic mineralogy, remanence carriers and nature of the remanence of the rocks.

PALAEOMAGNETIC RESULTS

NRM, susceptibility and Q-coefficient

The NRM directions of both intrusives and lavas were forming two groups of normal and reverse polarities far from the present dipolar direction (Fig. 3). Five sites of the granodiorites showed normal magnetizations and the other two sites showed reverse magnetizations. The lavas showed reverse

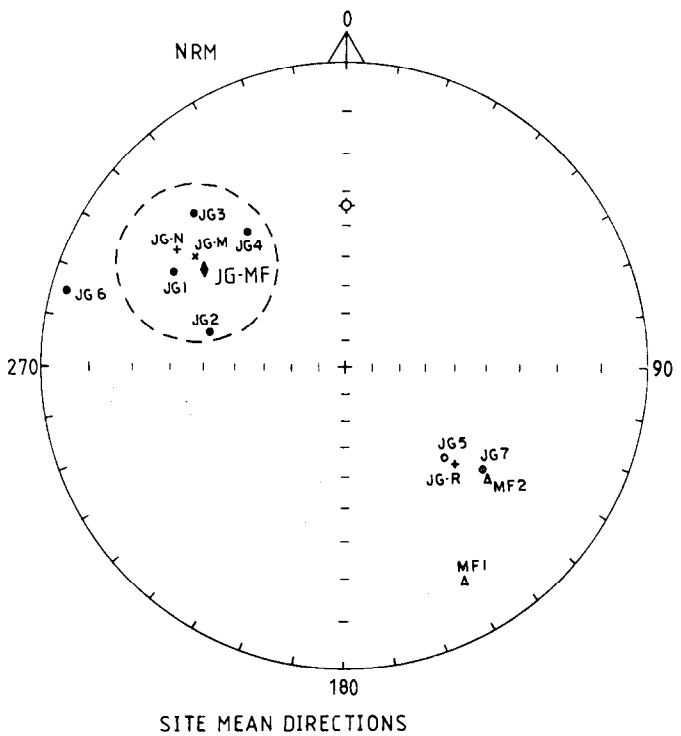


Fig. 3. Site mean NRM directions for intrusives (JG) and lavas (MF) with means for normal sites (JG-N), reverse sites (JG-R) and mixed polarity sites (JG-M). The overall mean (JG-MF) and α_{95} confidence limit is also indicated. The dipolar direction is ⊹, open symbols are negative inclinations and closed symbols are positive inclinations.

magnetizations which grouped not far from the granodiorite directions. The within-site scatter (as defined by the angle α_{95}) was lower than 20°, presenting similar characteristics for both normal and reverse directions (Table I). The NRM intensities and initial susceptibilities showed relatively greater variations for both within and between-sites; and it is possible to separate the sites in terms of these variations (Fig. 4). The higher NRM intensities of the granodiorites are associated with higher susceptibilities, which results in lower Q-coefficient values (NRM/xH, where NRM is the intensity of NRM, x is the susceptibility and H is the approximate magnetic field intensity for the sampling localities, ~0.4 Oe) than those of the lavas. Results are summarized in Table II.

AF and thermal demagnetization

Pilot specimens from sites JG1, JG3, JG4, JG5 and JG7 showed small directional movements about the NRM direction with the intensity decreasing with increasing temperature and alternating field (Fig. 5a, b). The AF pilot of site JG2 showed a somewhat irregular behaviour (Fig. 5b), distinct

TABLE I

Summary of initial data for the Jantetelco Granodiorites and Tepexco Volcanic Group *

Site	N	E	J_{NRM} (10^{-3} emu/cm^3)	Dec.	Inc.	α_{95}	k
1	4	4	0.185	299.1	25.0	20.0	22
2	5	5	0.025	283.6	41.3	10.3	56
3	6	14	0.434	314.9	20.4	7.0	92
4	6	13	0.202	323.7	32.8	2.6	653
5	6	12	1.009	132.4	−43.3	13.5	26
6	6	12	0.118	284.9	3.6	4.2	255
7	4	7	0.726	126.6	−32.4	15.1	38
Mean							
Normal	27	48	0.175	304.4	22.8	5.5	15
Reverse	10	19	0.750	131.1	−39.3	6.6	27
Mixed	37	67	0.264	306.1	27.6	4.6	15
All B = 7			—	298.6	34.3	30.4	5
Jantetelco Granodiorites							
1	6	12	0.034	151.2	−12.7	15.8	19
2	6	15	0.409	127.6	−29.9	14.4	23
All B = 2			—	140.1	−21.7	—	—
Tepexco Volcanic Group							
Overall mean							
B = 9				304.4	31.5	23.6	6

* Unit weight given to specimens to compute sample means; unit weight given to samples to compute site means.

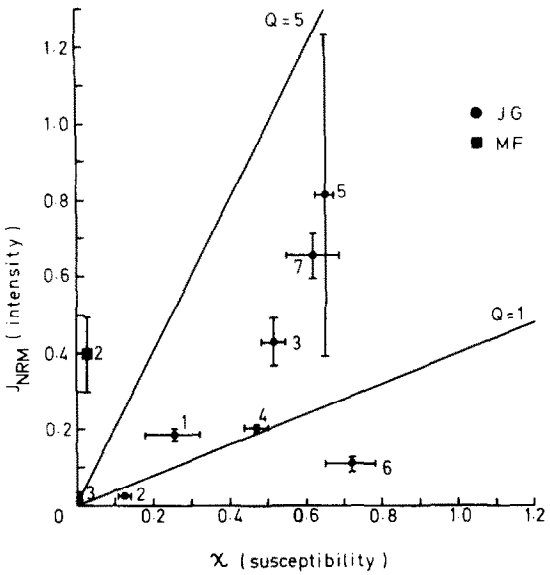


Fig. 4. Plot of NRM intensity (in 10^{-6} emu/cm^3) versus susceptibility (in 10^{-3} emu/cm^{3}Oe) for intrusives and lavas.

from the more stable behaviour of the thermal pilot and the remaining specimens of the site (Table III). Pilot specimens of site JG6 showed a systematic movement from a normal direction towards an intermediate reverse direction (Fig. 5a). This indicates the presence of at least 2 components of remanence

TABLE II

Summary of rock-magnetic data for the Jantetelco Granodiorites and Tepexco Volcanic Group *

Site	N	E	J_{NRM}	σ_J	X	σ_x	Q (H = 0.4)	MDF (Oe)	MDT (°C)
Jantetelco Granodiorites									
1	4	4	0.184	0.019	0.252	0.072	1.83	250	300
2	5	5	0.025	0.007	0.125	0.030	0.50	275	360
3	6	14	0.429	0.059	0.513	0.027	2.09	100	310
4	6	13	0.202	0.011	0.468	0.031	1.08	150	450
5	6	12	0.814	0.421	0.651	0.081	3.13	200	300
6	6	12	0.117	0.019	0.714	0.065	0.41	100	260
7	4	7	0.652	0.056	0.620	0.070	2.63	500	350
Tepexco Volcanic Group									
1	6	9	0.022	0.007	0.010	0.003	5.50	>900	>605
2	6	15	0.395	0.097	0.025	0.005	39.50	>900	>605

* Unit weight given to specimens. J_{NRM} given in 10^{-3} emu/cm^3 and X given in 10^{-3} emu/cm^3 Oe.

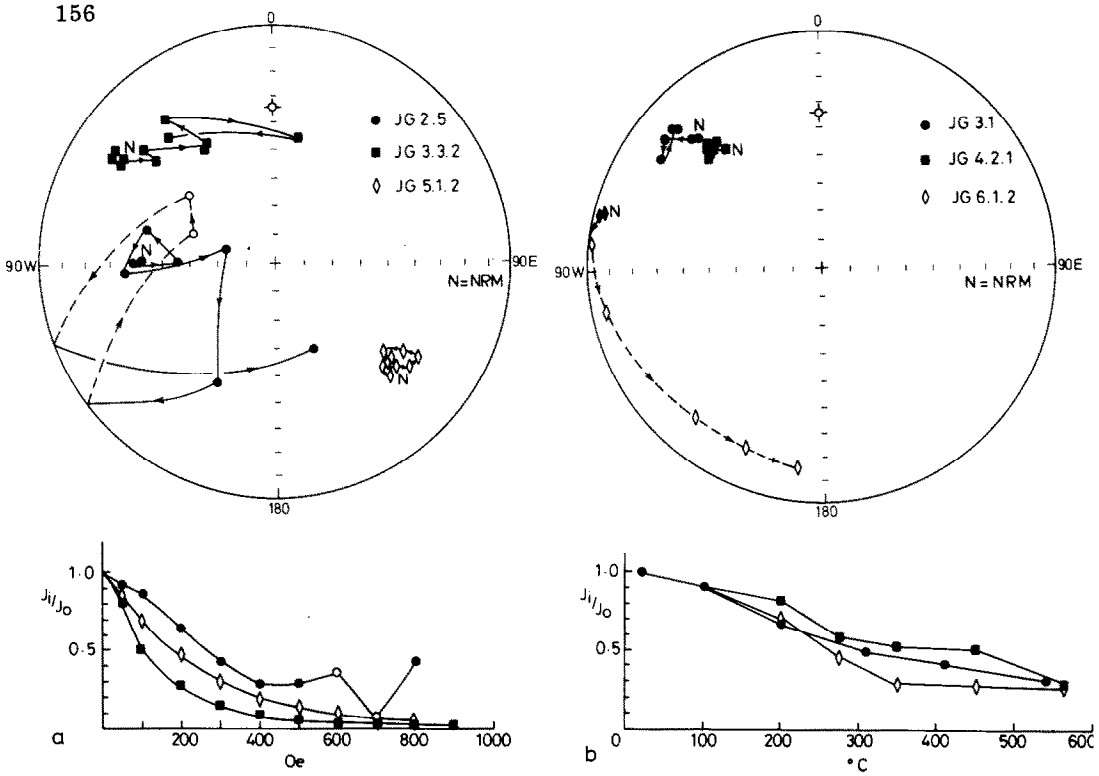


Fig. 5. Typical AF (a) and thermal (b) demagnetization results for the intrusives. Symbols as in Fig. 3.

with distinct coercivity and blocking temperature characteristics. Demagnetization of the remaining specimens of this site was carried out in two stages at 50—100° and 500—600°C or 100—200 and 800—900 Oe. This isolated a low coercivity—low blocking temperature magnetization (A-component) with normal polarity, and a second high coercivity—high blocking temperature magnetization (B-component) of intermediate polarity. Vector substraction analysis indicated a non-overlap region (initial linear segment in Zijderveld diagram of Fig. 6) in the spectra which permits definition of the A-component, but the spectrum of the B-component overlaps with either the spectrum of A or that of an unknown component, and it is not resolved in the ranges investigated. The mean direction obtained after the second demagnetization stage is:

N	E	Dec.	Inc.	α_{95}	k
6	12	190.4	—19.3	18.5	14.0

which is some 40° away from the directions given by stable end-points of lavas and intrusives (Table III). In an attempt to estimate the direction of the B-component, the method of converging remagnetization circles (Halls, 1976) was applied. The method may in certain cases be capable of isolating a given remanence direction, even when there is considerable overlap in the examined portions of the coercivity and blocking temperature spectra. Un-

TABLE III

Summary of palaeomagnetic data for the Jantetelco Granodiorites and Tepexco Volcanic Group

Site	N	E	Dec.	Inc.	α_{95}	k	c.s.d.	c.s.e.	Polarity	Lat. (°N)	Long. (°E)	A_{95}	k
Jantetelco Granodiorites													
1	4	4	300.6	20.4	6.2	222	5.4	2.7	N	32.1	171.9	5.0	344
2	5	5	281.6	38.1	9.4	68	9.9	4.4	N	17.0	188.9	9.6	64
3	6	14	310.6	17.3	3.6	350	4.3	1.8	N	41.2	166.7	3.4	396
4	6	13	319.8	31.8	2.7	637	3.2	1.3	N	51.8	175.6	2.5	705
5	6	12	133.0	−41.5	13.4	26	15.8	6.5	R	46.7	189.1	11.7	34
6	6	12	284.2	2.8	5.0	183	6.0	2.4	N	13.9	168.0	4.4	230
7	4	7	127.5	−31.9	10.5	77	9.2	4.6	R	40.5	178.5	10.6	77
Mean	37	—	302.6	26.6	5.9	17	19.7	3.2	M	35.4	176.6	5.2	22
	—	67	304.5	25.8	4.4	17	19.8	2.4	M	37.3	175.5	3.8	22
B = 7			302.2	26.9	14.2	19	18.5	7.0	M	35.1	176.8	12.4	25
Tepexco Volcanic Group													
1	6	9	110.2	−24.9	5.3	164	6.3	2.6	R	23.0	177.9	4.9	186
2	6	15	126.2	−30.4	10.9	39	13.0	5.3	R	39.1	177.7	11.7	34
Mean	12	—	118.0	−27.9	6.8	42	12.6	3.6	R	34.3	177.8	7.2	38
	—	24	121.3	−28.4	5.3	32	14.2	2.9	R	34.3	177.3	5.4	31
B = 2			118.0	−27.9					R	31.1	177.8		
Overall mean B = 9			301.2	27.1	10.7	24			M	33.7	176.3	9.5	30

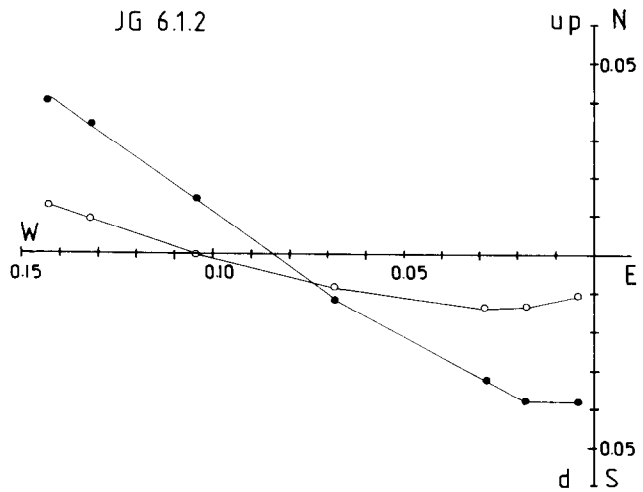


Fig. 6. Typical Zijderveld's (1967) demagnetization diagram. Open symbols (○) are successive total demagnetization vectors projected onto the horizontal plane, and closed symbols (●) are projected onto the vertical plane.

fortunately in this case the characteristics of the remagnetization circles obtained do not permit an unique isolation of the direction of the B-component. Nevertheless, the analysis indicates a reversely magnetized direction of

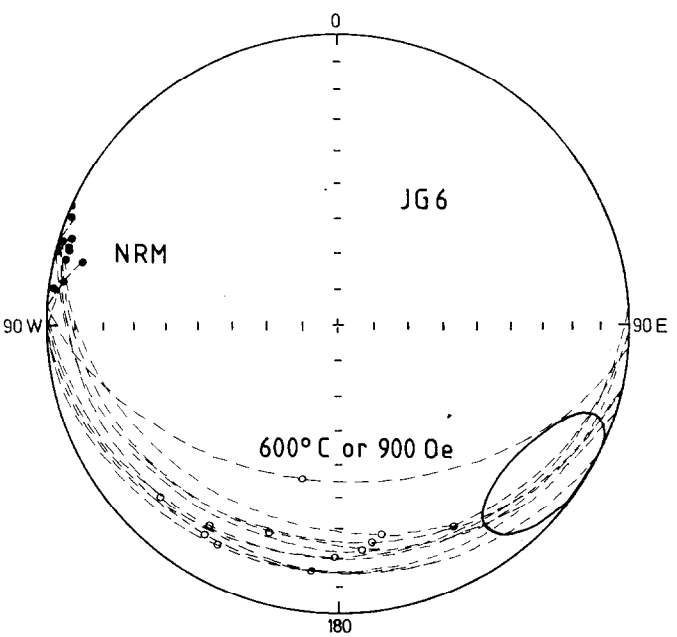


Fig. 7. Remagnetization circles (Halls, 1976) for site JG6. Each curve represents a specimen. Only the initial and end points are indicated.

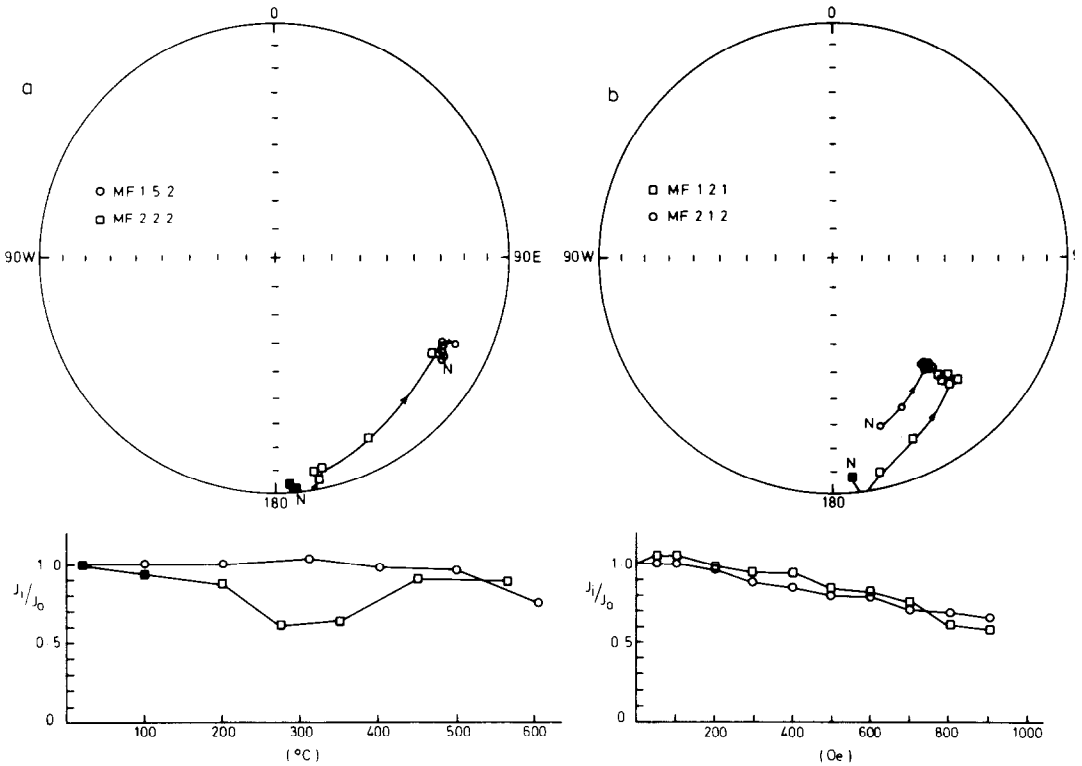


Fig. 8. Typical AF (a) and thermal (b) demagnetization results for the lavas. Symbols as in Fig. 3.

some 110—140° declination and some —6° to —18° inclination (Fig. 7). The inclinations are shallower than those isolated from the remaining sites of intrusives and lavas (Table III). Because of the large uncertainties associated with this analysis, the results were not considered for the final statistical analysis.

In the case of the lava flows, most samples showed a secondary magnetization of possible recent viscous nature which was eliminated by the treatments, and stable magnetizations from well defined end points were isolated (Fig. 8a, b).

Results from all demagnetized samples were averaged by giving unit weight to specimens to compute sample means and then by giving unit weight to those means to compute site means. Overall means for the granodiorites and lavas were computed giving unit weight to specimens, samples and sites, which gave very similar results. Finally a combined mean direction was calculated giving unit weight to all sites. These results and corresponding pole positions are summarized in Table III.

Rock-magnetic results

The behaviour of all pilot specimens during AF and thermal demagnetization is indicative of an intermediate coercivity mineral, probably fine-grained magnetite, with a small amount of what is probably hematite. Both magnetic carriers occur in all samples, but those of site JG6 recorded the same direc-

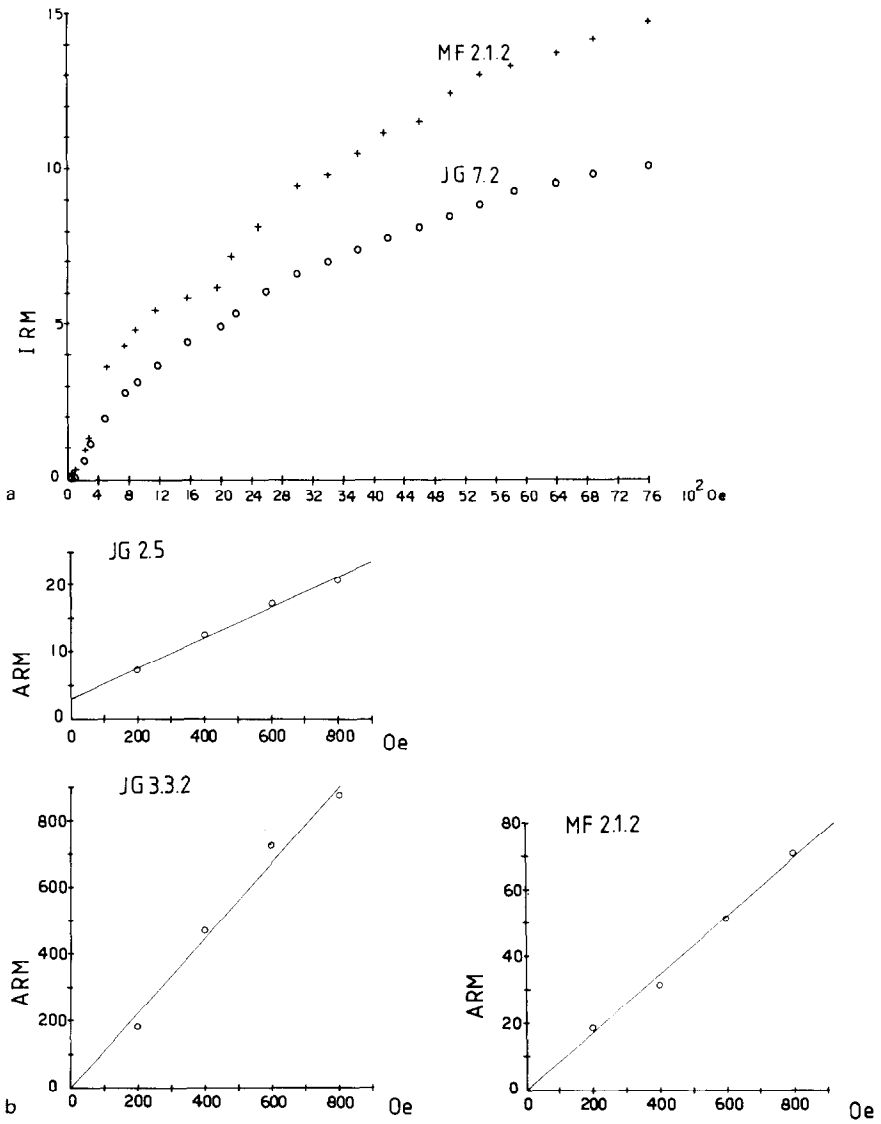


Fig. 9. IRM acquisition curves (in 10^{-3} emu/cm^3) (a) and ARM acquisition curves (in 10^{-6} emu/cm^3) (b) for lavas and intrusives.

tion of remanence. The IRM acquisition curves (Fig. 9a) did not show complete signs of saturation in fields up to 7.6 K Oe, although the curves, particularly those of the granodiorites, started to level off in fields of some 4 K Oe. These curves may indicate magnetite with a contribution from a high coercivity mineral such as hematite or goethite (Dunlop, 1972). The lavas probably contain a relatively larger amount of the high coercivity mineral, which agrees with the results of the thermal pilots which indicate the presence of high blocking temperature minerals. ARM acquisition curves (Fig. 9b) showed almost linear relationships with increasing alternating field, which in most cases went through the origin. Such relationships might be related with the magnetic stability since linear relationships going through the origin seem to correlate with high directional stability (Urrutia-Fucugauchi, 1980). AF demagnetization of the maximum IRM and ARM showed slightly more rapid decays with increasing fields than that of NRM (Fig. 10), which may indicate that the remanences reside in single domain grains (Lowrie and Fuller, 1971). The median destructive field (MDF), or field required to reduce the NRM intensity to half its initial value, ranges from 100 to 275 Oe for the granodiorites, except for site JG7 with values of about 500 Oe, which may indicate the presence of pseudo-single domain and single domain magnetic grains (Parry, 1965). These MDF values are lower than those of the lavas,

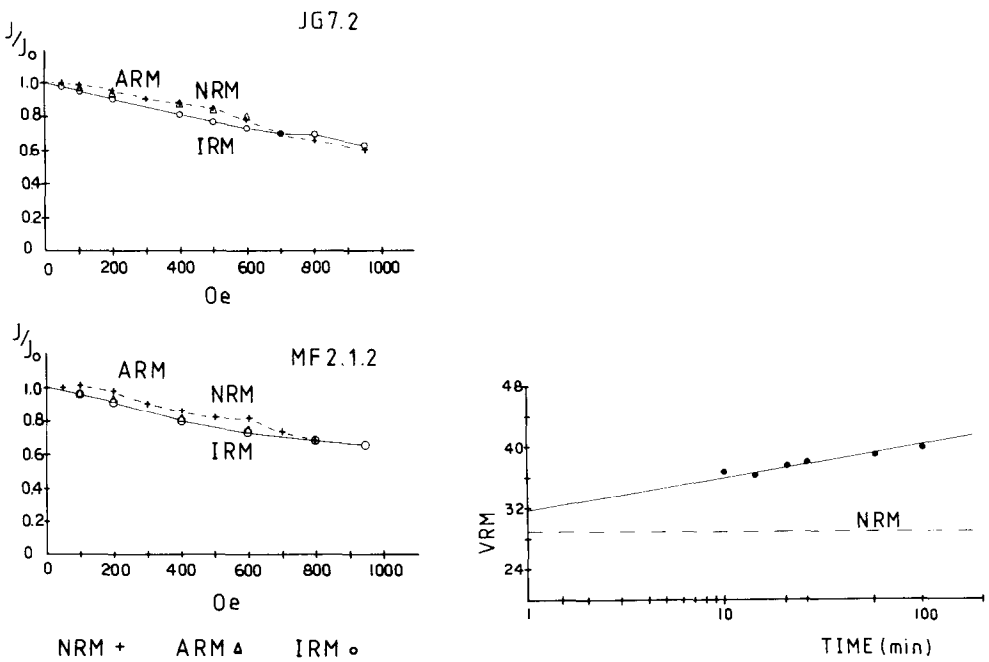


Fig. 10. AF demagnetization curves for NRM, ARM, and IRM.

Fig. 11. VRM build-up for an intrusive specimen of site JG2 (units in 10^{-6}emu/cm^3); applied field 0.5 Oe.

which are in excess of the maximum AF peak field available. The median destructive temperature (MDT), or temperature required to reduce the NRM intensity to half of its initial value, ranges from about 260—450° for the granodiorites, again lower than those of the lavas which are in excess of 600°C (Table II). The high MDF and MDT values of the lavas indicate the presence of high coercivity—high blocking temperature minerals as remanence carriers.

One specimen from each site was exposed to a low magnetic field (0.5 Oe) to study the acquisition of viscous remanence (VRM) during relatively short intervals (up to 100 min). Only the specimen from site JG2 showed observable and consistent variations which gave a linear relationship with the logarithm of time of exposure (Fig. 11). Larger variations were also observed during NRM measurements for the pilot specimen JG2.5, which may explain the erratic behaviour observed during AF demagnetization (Fig. 5b).

DISCUSSION

The geomagnetic field at the time of remanence acquisition of the lavas was of reverse polarity. Two sites (JG5 and JG7) of the granodiorites present

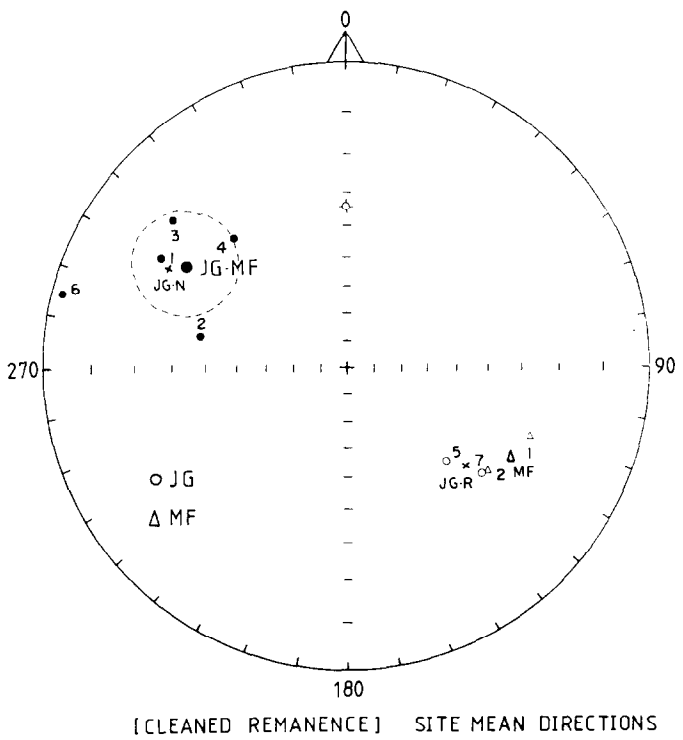


Fig. 12. Site mean directions after AF and thermal demagnetization for all sites. Overall mean for intrusives and lavas with the α_{95} confidence limit is given by JG-MF. Symbols as in Fig. 3.

TABLE IV

Comparison with results from northern Mexico and North America

a. Summary of palaeomagnetic pole positions

	Time span	No. of studies	Lat. (N)	Long. (E)	A_{95}	Reference
1. N.A. mode pole	7—21 M	4	83	203	8	Van Alstine and De Boer (1978)
2. N.A. mean pole	7—21 M	4	86	195	8	Van Alstine and De Boer (1978)
3. N.A. pole	Late Miocene	14	87	167	5	Irving (1979)
4. N.A. pole	Oligo—Miocene	21	87	162	4	Irving (1979)
5. Northern Mexico	Miocene	5	82	148	7	Urrutia-Fucugauchi (1979b)
6. Pole this study	Miocene	9 sites	34	176	10	

b. Summary of expected mean palaeomagnetic directions and palaeolatitudes

Calculated from	Dec.	Inc.	Palaeolatitude	Lat.-differences with 6
1.	353.6	39.3	22.2	8
2.	356.1	36.4	20.2	6
3.	356.8	33.7	18.4	4
4.	356.9	33.3	18.2	4
5.	352.4	28.8	15.4	1
6. This study	301.2	27.1	14.3	(present lat. 18.67°N)

reverse magnetizations, which are very close to those of the lavas (Fig. 12). Since there is no evidence of remagnetization for the lavas and granodiorites, the field at the time of remanence acquisition of the granodiorites might have been similar to that present during the eruption of the lavas. Samples from site JG6 seem to have recorded a polarity transition (marked by 2 remanence components). The remaining 4 sites of the granodiorite, corresponding to the largest body, present normal polarity. Thus, the cooling of the granodiorite spanned a polarity transition of the geomagnetic field. This may indicate that the emplacement and cooling of the different bodies occurred at slightly different times, or that the cooling of the smaller bodies was faster than that of the larger central body, or that if the bodies belong to a single larger body, sites JG5-7 were nearer to the edges of the body and sites JG1-4 were nearer to the center.

The mean palaeomagnetic pole positions derived for the granodiorites and lavas (Table III) are of low latitude and define an overall mean palaeomagnetic pole at 33.7°N and 176.3°E. Comparison with mean Miocene poles for Mexico (Fig. 13) and the stable part of the United States and Canada (Table IVa) shows angular differences in excess of 50°. These large discrepancies are unlikely to be due to secular variation effects. The results from the lavas and granodiorites are fairly consistent (Table III and Fig. 13) and present both normal and reverse magnetizations, forming two almost opposite groups. To investigate the secular variation characteristics recorded by the intrusives and lavas, the standard deviation due to the ancient geomagnetic field (S_a) was calculated from:

$$S_a^2 = S_b^2 - S_1^2$$

where S_b represents the between-site dispersion, which is estimated from the population of pole positions (Cox, 1962), and S_1 represents the standard deviation due to local effects, which is taken as 1.8° (Doell and Cox, 1963). The value estimated is about 14.8°, with upper and lower 95% confidence limits (Cox, 1969) of 21.4° and 11.3°, respectively. This result agrees well with the expected dispersion predicted by model A (Irving and Ward, 1964), and is within the range predicted by other secular variation models (McElhinny and Merrill, 1975; Isaacson and Heinrichs, 1976). Assuming that non-dipole components during the Miocene were not significantly different from those of the present and that this study has successfully isolated the palaeomagnetic field, then the results may indicate the effects of large scale local or regional tectonic movements. This suggests that this part of Mexico has rotated counterclockwise with respect to northern Mexico and North America since Miocene times.

Because much of the geological—geophysical evidence concerning the past relative motions between the Farallon, Pacific and North American plates has disappeared due to plate subduction, investigations are largely dependent upon evidence from the Pacific and North American plates (Atwater, 1970).

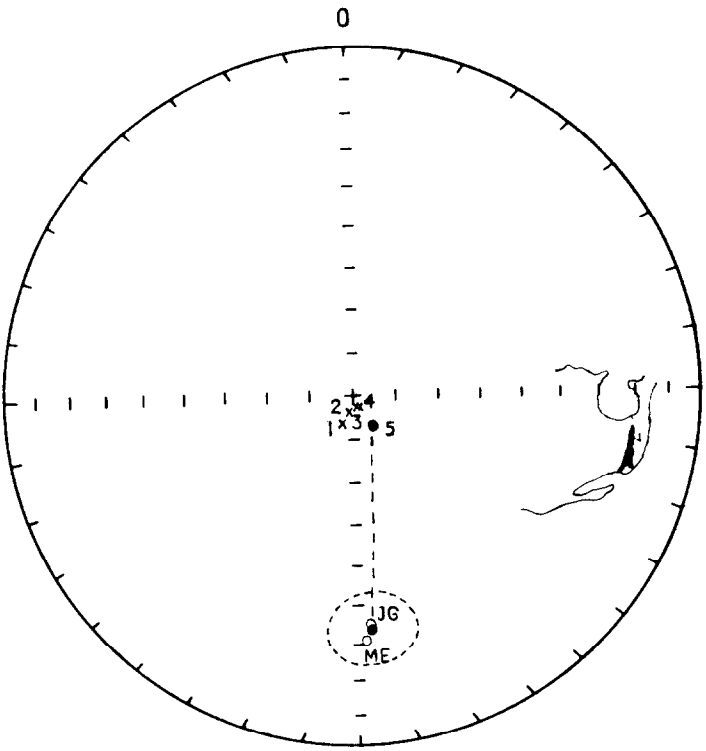


Fig. 13. Comparison of the mean pole positions for the granodiorites (JG), lavas (MF) and combined result (closed circle) with mean pole position for North America (x) and northern Mexico (●). Numbers refer to Table IVa. The sampling locality is indicated by Δ. The α_{95} confidence limit for the combined mean pole position for the granodiorites and lavas is included, and the angular displacement with respect to the mean Miocene pole for northern Mexico is also indicated. The black area within Mexico represents the Mexican Volcanic belt.

The panorama during the Tertiary is that of a large scale microplate reorganization, where the most significant events might have been related to the subduction of the Farallon rise and subsequent fragmentation of the Farallon plate (Atwater, 1970: Atwater and Molnar, 1973). This resulted in the separation of the Baja California peninsula away from the mainland and the formation of the Gulf of California by sea-floor spreading and transform fault movements (Larson et al., 1968; Moore, 1973). The plate boundary reorganization resulted in important changes in the character and location of magmatism related to plate subduction. McDowell and Keizer (1977) have indicated that voluminous rhyolitic activity occurred 32—23 Ma ago in western Mexico (Sierra Madre Occidental) and that intense volcanic activity started in Baja California by 20 Ma. Activity in the Mexican Volcanic Belt also started during the Late Oligocene, with a major episode of activity during the Miocene (Mooser, 1972). Several authors (e.g. Mooser and Maldon-

ado-Koerdell, 1961; Phillips, 1964; Mooser, 1968, 1972; Gastil and Jensky, 1973; Pal and Urrutia-Fucugauchi, 1977) have indicated the presence of a large crustal discontinuity beneath the MVB. Gastil and Jensky (1973) have suggested that this represents a regional fault system which has undergone right lateral strike slip movements involving some 175 km displacement during Late Cretaceous or Early Tertiary times and some further 260 km during the Miocene or Pliocene time. To investigate the nature of tectonic movements involved, the paleomagnetic directions expected from data of northern Mexico and North America were calculated (Table IV). The calculated declinations are some 50—55° displaced clockwise from the observed declination, whereas the inclinations are only some 1—8° steeper than the observed inclination. These differences might be accounted for by some 50° counterclockwise rotation about a vertical axis since the emplacement of lavas and intrusives. Freund (1974) has suggested that rotation of crustal blocks in areas subject to shear stresses is a common process. Beck (1976) interpreted anomalous palaeomagnetic data from several areas in the southwestern United States in terms of rotation of small rigid microcontinental blocks and northward translations, the so-called "ball-bearing hypothesis" (see fig. 4 of Beck, 1976). A possible mechanism is then penetrative shear associated with large-scale strike slip motion distributed throughout the sedimentary and volcanic formations, where the area rotated about a nearby vertical axis. Alternatively, the results might be explained in terms of the formation of a pull-apart basin. Crowell (1974) has suggested that if a strike slip fault develops a double bend, continued motion along the fault will produce separation of the blocks along the oblique segment of the fault resulting in the formation of a pull-apart basin. Greenhaus and Cox (1979) have suggested that this process might be accompanied by rotation of loosened crustal blocks within the basin (see fig. 7 of Greenhaus and Cox, 1979). The dimensions of the basin are related to the initial length of overlap of the faults, their separation, and the amount of strike slip displacement.

The broad fault zone along the trace of the MVB may have provided an environment favorable to the arc magmatism product of the subduction of the Cocos plate at the Middle America trench. Thus, the zone of crustal weakness formed may have acted as an efficient structural control for the magmas.

CONCLUSIONS

The present study provides support for ideas about tectonic movements along the trace of the MVB. The palaeomagnetic results are interpreted in terms of a tectonic rotation of a crustal block about a vertical axis located somewhere nearby, this possibly in response to regional lateral shear. More detailed palaeomagnetic sampling is necessary to give a more complete picture of the tectonic evolution. We may then be able to find support for the tectonic rotation proposed, and evidence concerning the number and size of

rotated blocks, which may comprise either minor portions of the area or may involve the whole of central Mexico.

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